

Thermochemistry

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Enthalpy

The enthalpy of a system is defined as the sum of the internal energy and the product of its pressure and volume.

That is, $H = E + PV$

Where E is the internal energy, P is the pressure and V is the volume of the system. It is also called Heat content.

If ΔV be the change in volume in case of a reaction at constant temperature and pressure, the thermal effect observed will be the sum of the change in internal energy (ΔE) and the work done in expansion or contraction, that is

$$\Delta H = \Delta E + P \times \Delta V$$

Exothermic reaction

The reactions which are accompanied by the evolution of heat energy are called exothermic reactions.

The evolved heat is expressed by adding it with the products or as ΔH . The value of ΔH will be negative in the reaction.



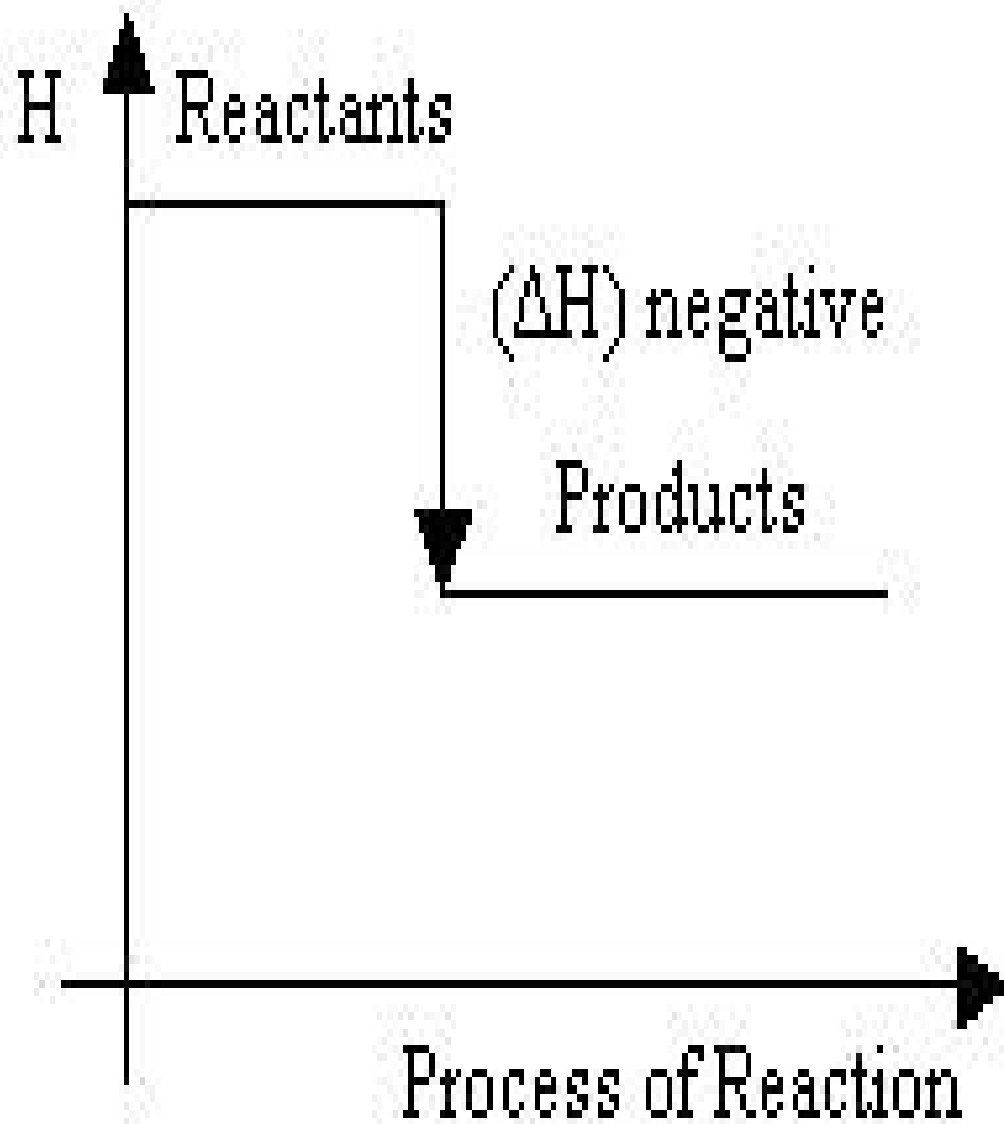
Endothermic reaction

The reactions which are accompanied by absorption of heat are called endothermic reaction.

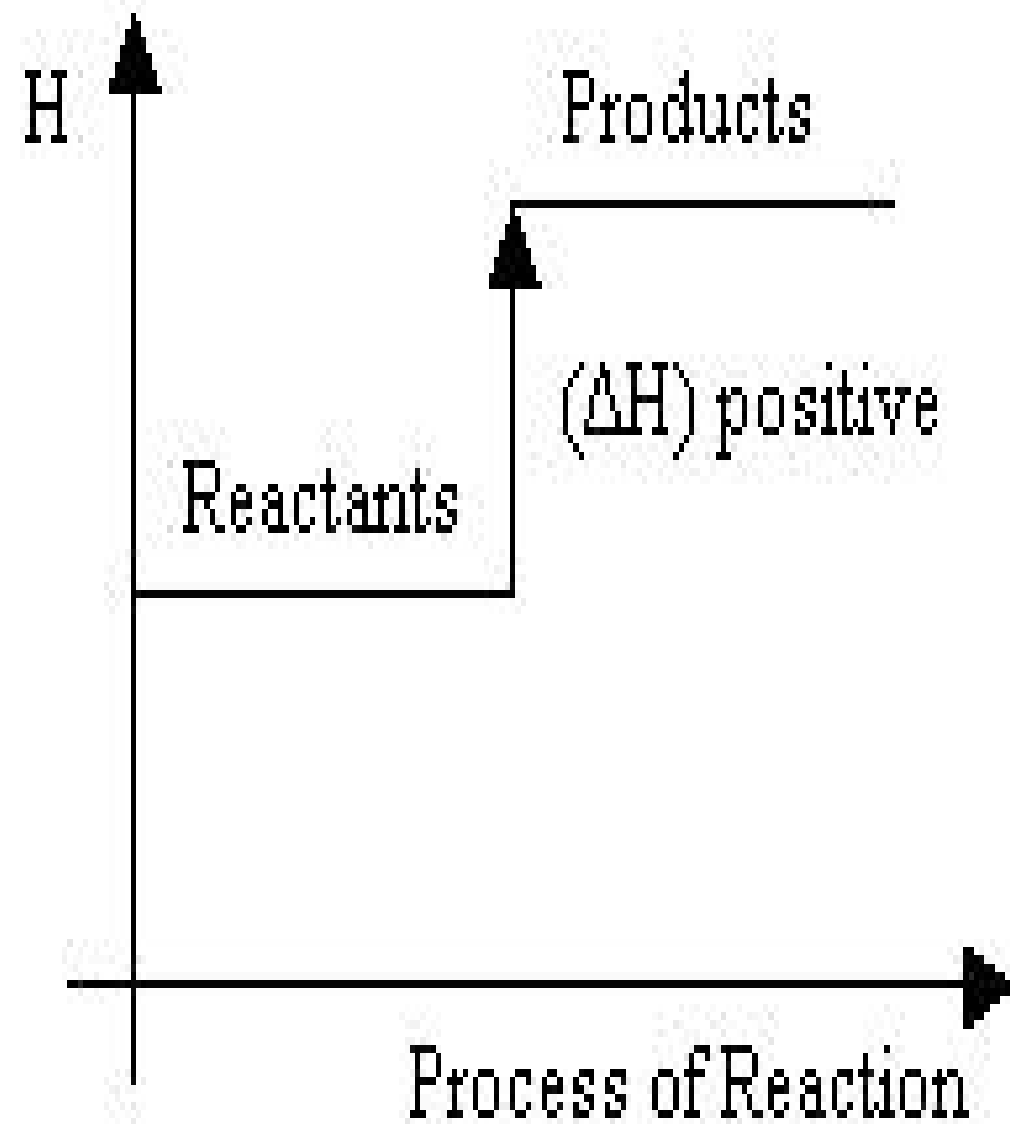
The value of ΔH will be positive in endothermic reactions.



Exothermic



Endothermic



Heat of reaction

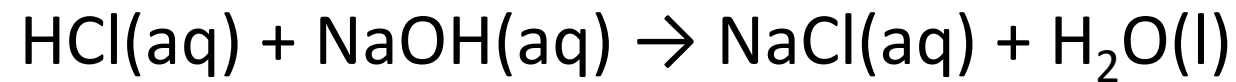
The amount of heat absorbed or evolved in a reaction when the number of moles of reactants as represented by the balanced chemical equation change completely into the products



The heat change accompanying a reaction taking place at 298 K and one atm is called the standard heat change or standard enthalpy change. It is denoted by ΔH°

Heat of Neutralisation

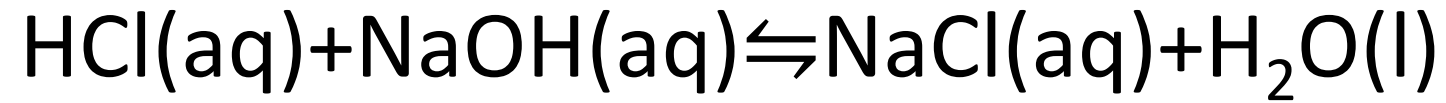
The change in enthalpy of the system when one gram equivalent of an acid is neutralised by one gram equivalent of a base or vice versa in dilute solution.



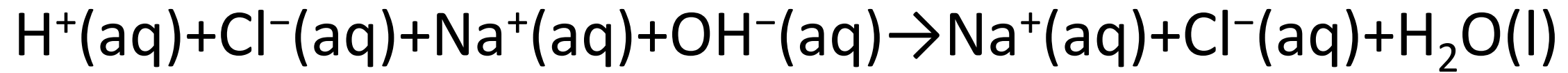
- **Heat of neutralisation of strong acid and strong base is constant**

The strong acids and strong bases are almost completely ionized in dilute aqueous solutions. The neutralization of strong acid and strong bases simply involve the combination of H^+ ions (from acid) and OH^- ions (from base) to form water molecules. It has been found experimentally that when 1 mole of water is formed by the neutralization of 1 mole of H^+ ions and 1 mol of OH^- in aqueous solutions, 57.1 kJ of energy is released.

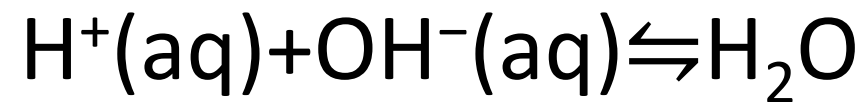
Consider the reaction between HCl and NaOH in water:



This can be written in terms of the ion

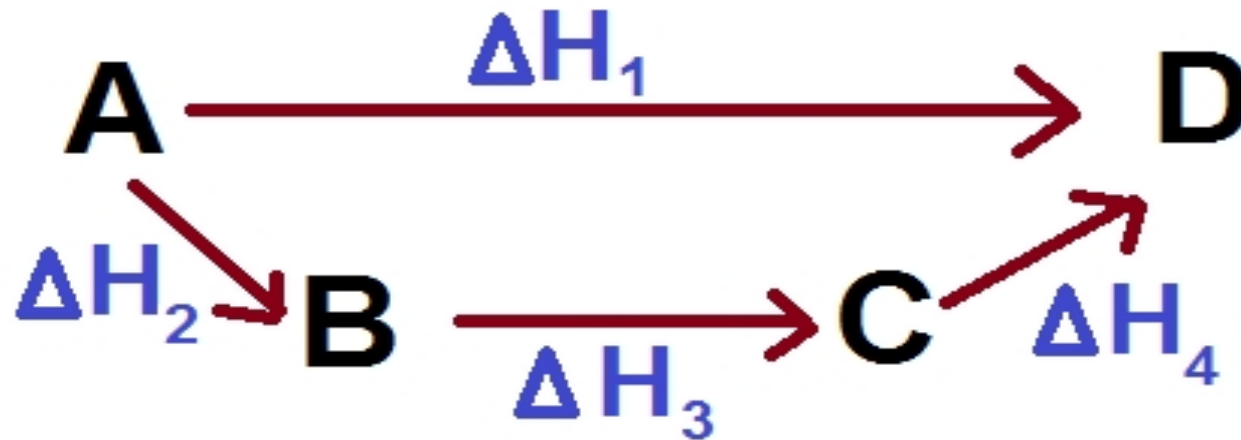


When the spectator ions are removed, the net ionic equation shows the H^+ and OH^- ions forming water in a strong acid, strong base reaction:



Hess's Law of constant heat summation

If a chemical reaction can take place by two or more different ways the initial and final conditions are the same, the total enthalpy change is the same.



$$\Delta H_1 = \Delta H_2 + \Delta H_3 + \Delta H_4$$

1st way:

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Hess's Law

(statement, Illustration, application, Problems)

Hess's Law may be stated as:

If a chemical change can be made to take place in two or more different ways whether in one step or two or more steps, the amount of total heat change is same no matter by which method the change is brought about.

1st way :



2nd way :



SOLVED PROBLEM 1. Calculate the heat of formation of potassium hydroxide from the following data.

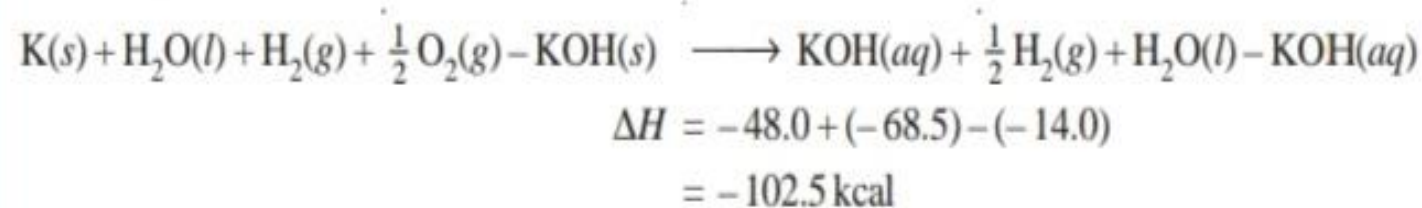


SOLUTION

We should aim at finding the value of ΔH for the equation, which is the heat of formation.



Adding equations (i) and (ii) and subtracting equation (iii) will give



Thus heat of formation of KOH is **-102.5 kcal**

SOLVED PROBLEM 2. The heat of combustion of ethyl alcohol is -330 kcal. If the heat of formation of $\text{CO}_2(g)$ and $\text{H}_2\text{O}(l)$ be -94.3 kcal and -68.5 kcal respectively, calculate the heat of formation of ethyl alcohol.

SOLUTION

We are given

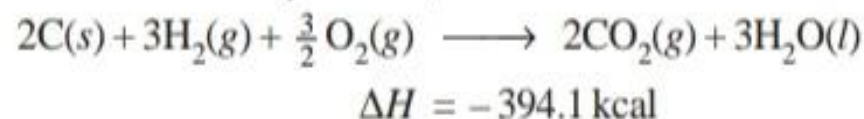


We have to manipulate these equations so as to get the required equation :

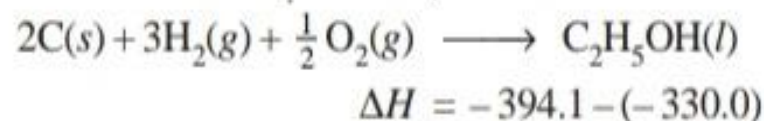


where ΔH is the heat of formation of ethyl alcohol.

Multiplying equation (b) by 2 and equation (c) by 3 and adding up these, we get

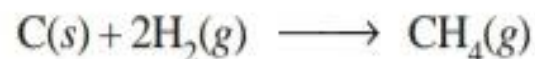


Subtracting equation (c) from the above equation, we have



Thus the heat of formation of ethyl alcohol is **-64.1 kcal.**

SOLVED PROBLEM 3. Determine ΔH of the reaction

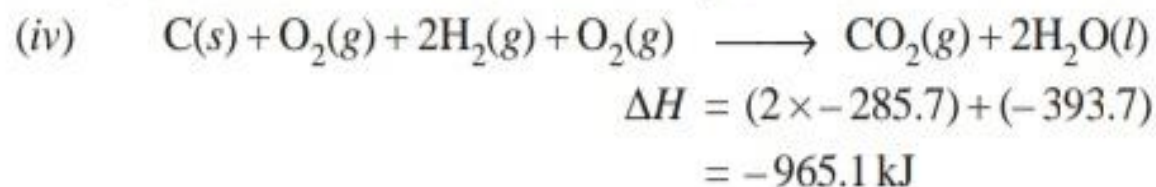


from the following data :



SOLUTION

Multiplying equation (ii) by 2 and adding to equation (i), we get



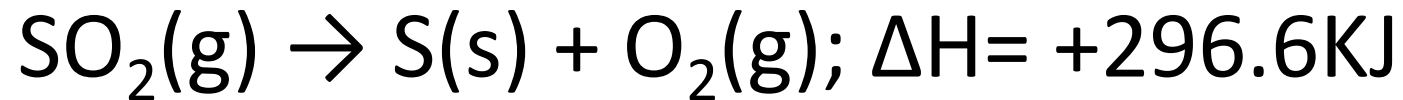
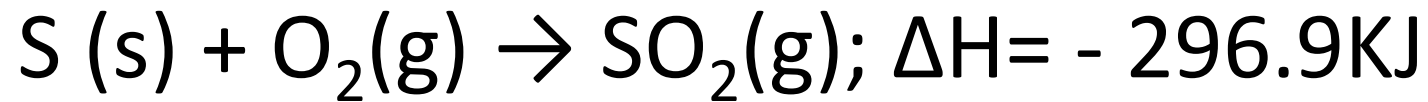
On subtracting equation (iii) from equation (iv), we get



Thus the heat of formation of methane is **-74.8 kJ**.

Lavoisier and laplace law

The heat changes associated with a chemical process proceeding in a direction are exactly equal but opposite in sign for the reverse reaction.

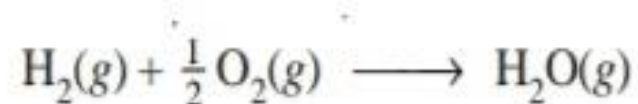


Bond Energy

- The bond energy is defined as the average amount of energy required to break all bonds of a particular type in one mole of the substance.

Bond	Bond Energy
Cl-Cl	243 kJ mol ⁻¹
O=O	499.0 kJ mol ⁻¹
C-H	414.0 kJ mol ⁻¹
O-H	460.0 kJ mol ⁻¹

SOLVED PROBLEM 1. Given that energies for H–H, O=O and O–H bonds are 104, 118 and 111 kcal mol⁻¹ respectively, calculate the heat of the reaction



SOLUTION

In this reaction, two O–H bonds are formed and one H–H bond is broken. Therefore we can write for ΔH

$$\begin{aligned}\Delta H &= -2 \Delta H (\text{O} - \text{H}) + \Delta H (\text{H} - \text{H}) + \frac{1}{2} \Delta H (\text{O} = \text{O}) \\ &= (-2 \times 111) + 104 + \left(\frac{1}{2} \times 118\right) \\ &= -222 + 104 + 59 \\ &= -59 \text{ kcal mol}^{-1}\end{aligned}$$

$\therefore$ The heat of the given reaction is **– 59.0 kcal mol⁻¹**